bigt.ai

P2P Settlement — Wallet-Native CNY Swap

What this is. A peer-to-peer swap between crypto on **Bigtangle L0** and CNY fiat (WeChat Pay / Alipay / bank). The seller escrows tokens in a **2-of-3 P2SH address **(seller, buyer, engine); the buyer pays the seller** directly** in their own payment app; the seller confirms receipt on their own statement and the escrow releases. Every irreversible step is signed with the wallet's own PQ key, and the engine can only ever **co-sign** a spend — it never holds your funds and never touches fiat.

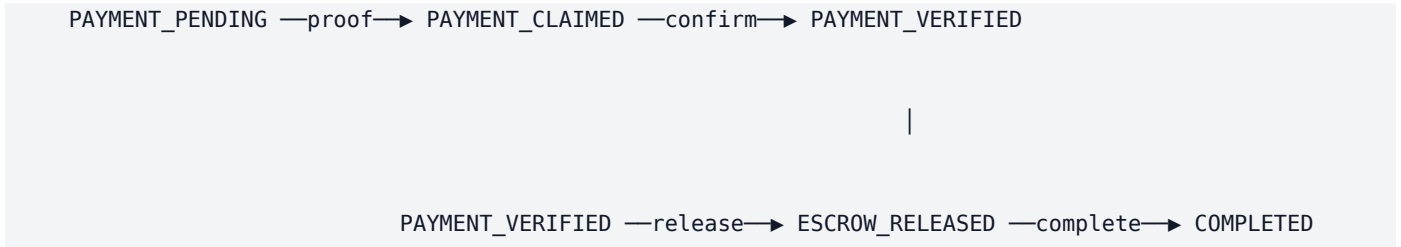
Where. The **P2P** screen in the wallet (sidebar → Trade → P2P). Everything below is captured from that screen on the local demo build.

Demo vs. live. The screenshots come from the demo build: the engine runs with a memory store and no chain check, so the lock/release legs use synthetic transaction hashes. The state machine, the signatures, the append-only event log and the on-chain audit anchor are the real thing; only the L0 broadcast is stubbed.

Seller confirmation is the fiat signal. There is no collection API for a personal WeChat/Alipay/bank account, so there is no machine-verifiable payment event to wait on. The seller checks **amount + remark + time + 流水号 in their own app** — a screenshot never counts — and confirms. Design details live in `docs/p2p.md`.

The flow at a glance

```
ACTIVE ──match──▶ MATCHED ──lock──▶ ESCROW_LOCKED ──instructions──▶ PAYMENT_PENDING
                                                                          |
```



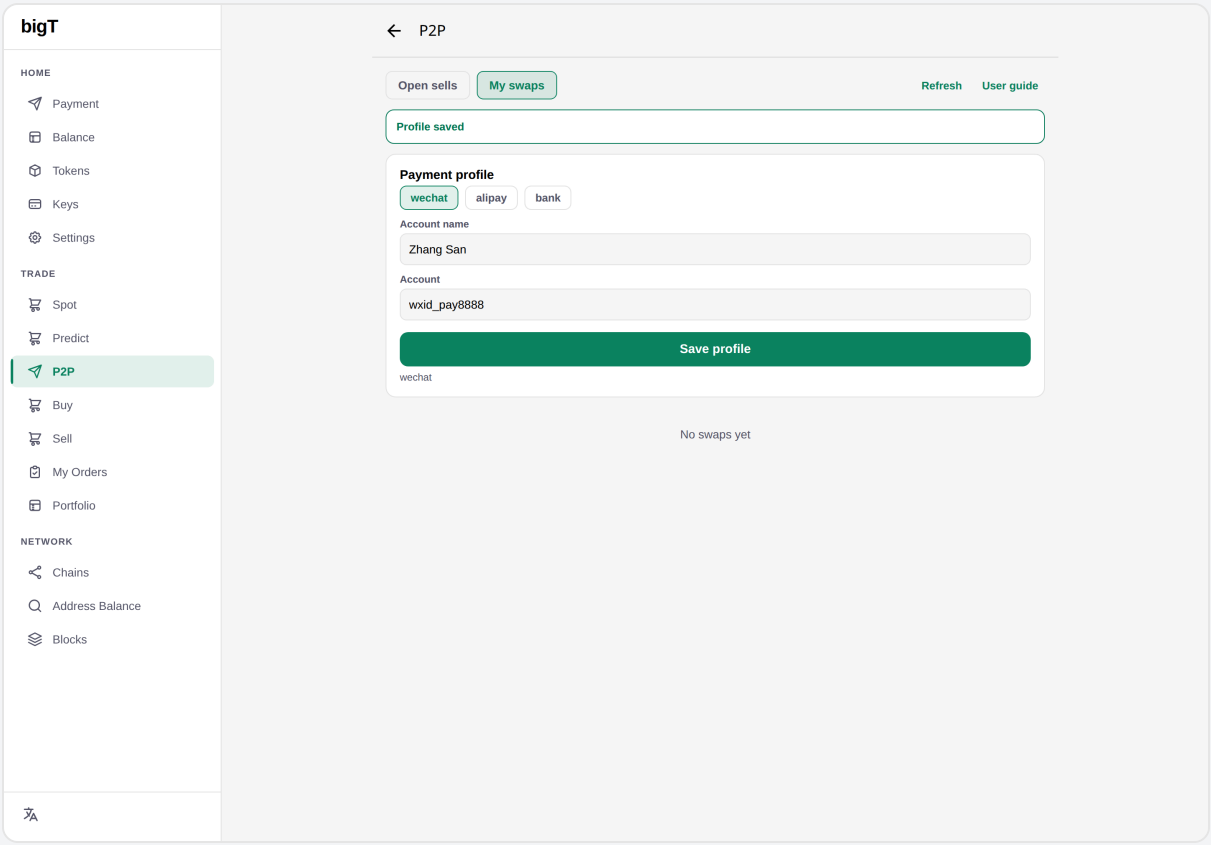
#	Step	Who acts	On-chain / engine
1	Seller saves a CNY collection profile	seller	profile stored server-side, never in the book
2	Seller lists a signed sell order	seller	order stored, not funded
3	Buyer matches (receive address)	buyer	`swapId` created, escrow address derived
4	Seller funds the 2-of-3 escrow, engine proves the lock	seller + engine	`ESCROW_LOCKED` on L0
5	Buyer pulls the payment instructions	buyer	`PAYMENT_PENDING` (+ per-swap remark)
6	Buyer pays in their own app, claims with the 流水号	buyer	`PAYMENT_CLAIMED`
7	Seller checks their own statement and confirms	seller	`PAYMENT_VERIFIED`
8	Engine releases the tokens, swap completes	engine	`ESCROW_RELEASED` → `COMPLETED`

Every transition is anchored as a `social.p2p-swap` record on the L1-SOCIAL chain, so the whole lifecycle is publicly auditable without exposing either party's account details.

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1. The seller saves a collection profile

On the **My swaps** tab the seller picks a rail (WeChat, Alipay or bank) and enters the 收款 name and account. The profile is stored server-side under the seller's DID and is **not** part of the public order book — it is revealed only to the matched buyer by the instructions call.

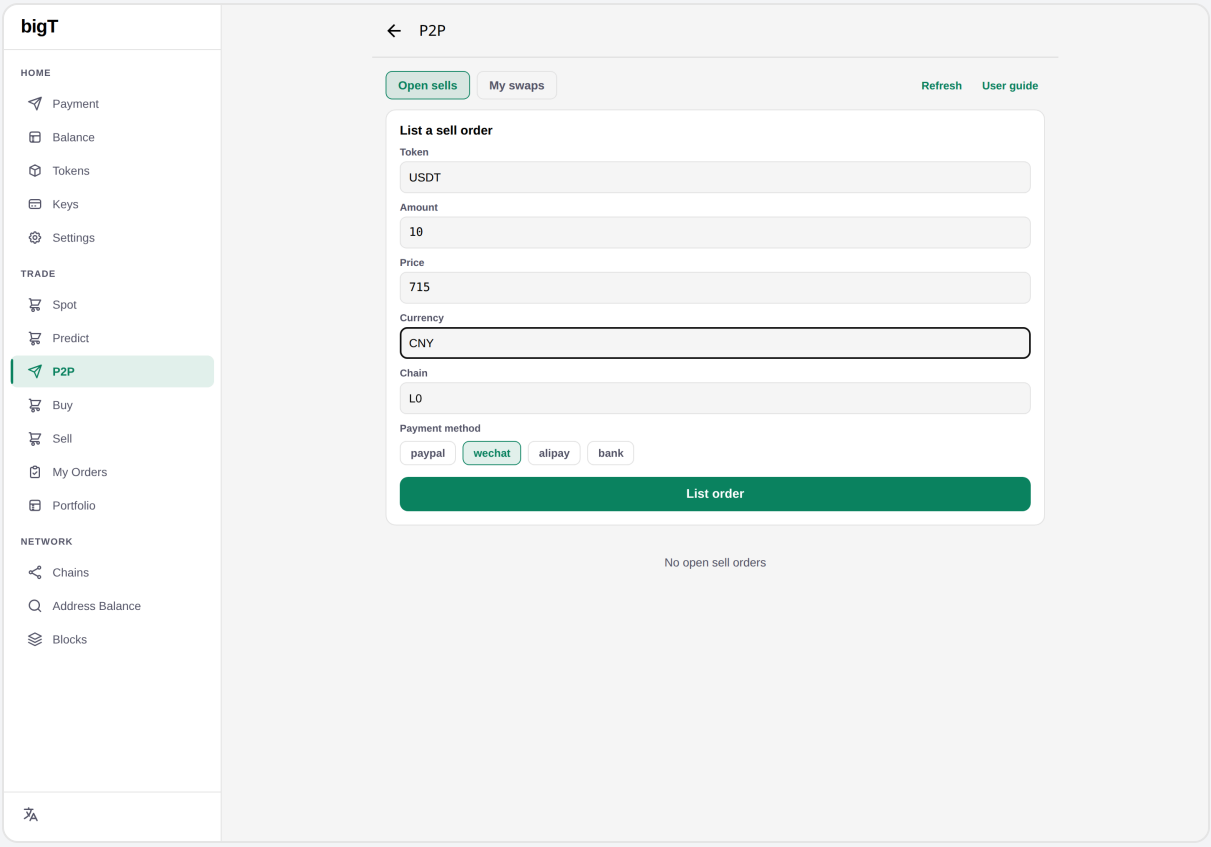


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2. The seller lists a sell order

On the **Open sells** tab, fill in the order: the token and amount you give, the CNY price you want, the chain the tokens sit on, and the **payment method** (wechat / alipay / bank). Selecting a CNY rail switches the currency to CNY.

Terms are locked into the order when you list it, so this is signed with your wallet key:

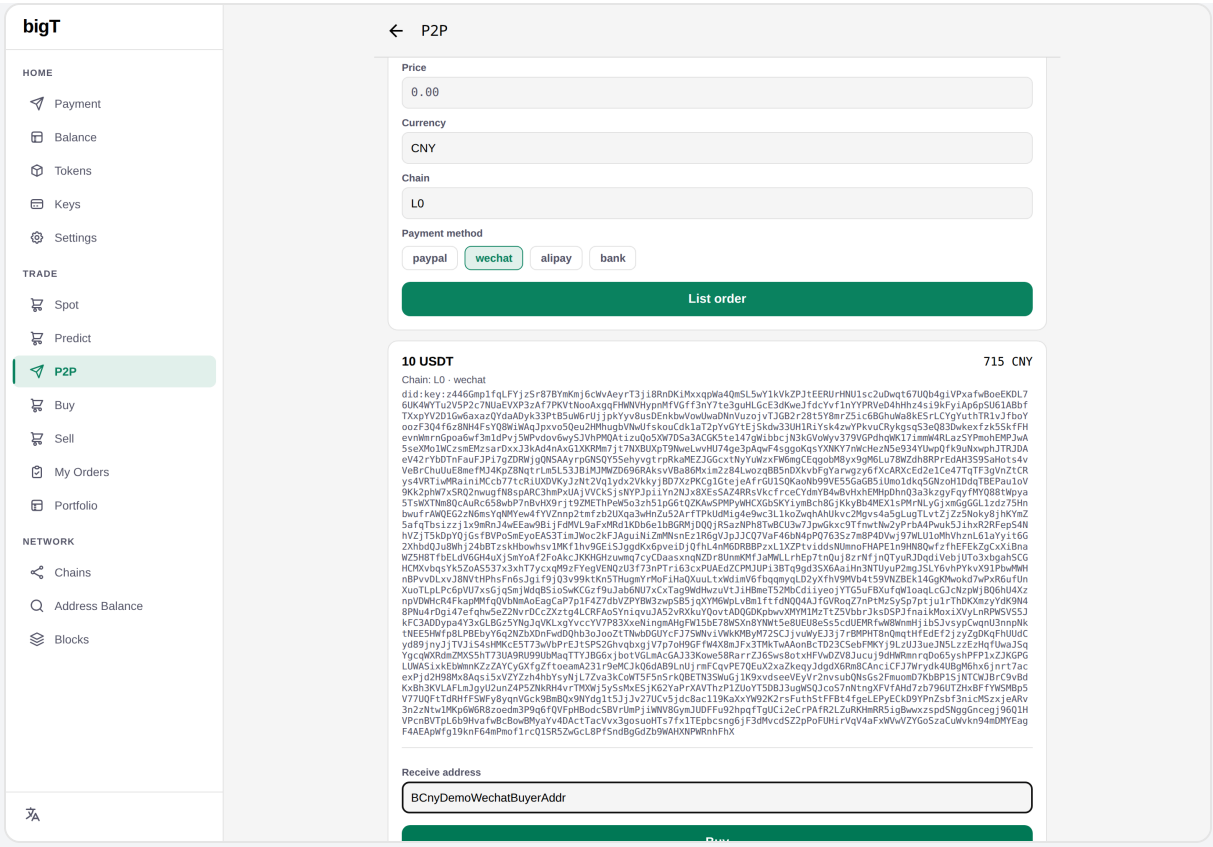


Once listed, the order is `ACTIVE` in the engine and appears in the public order book (it carries no personal details).

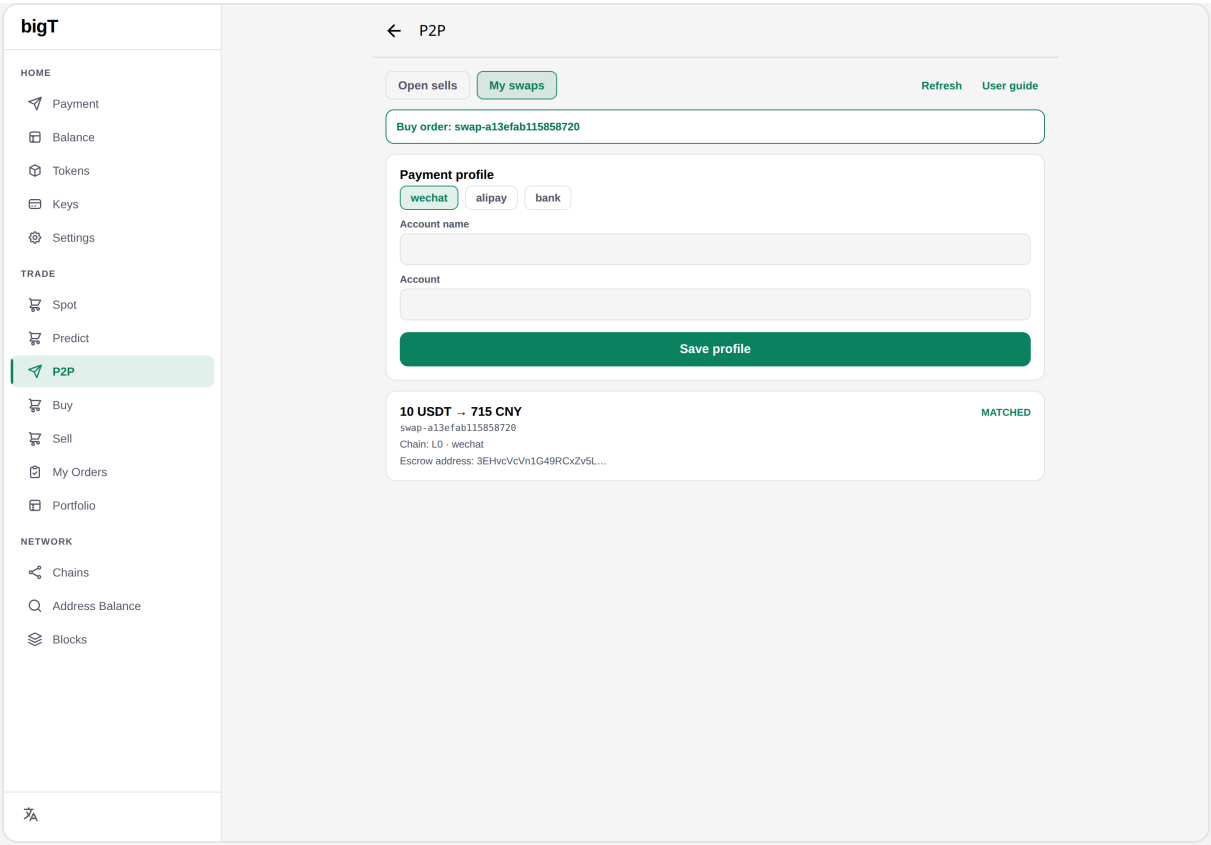
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3. The buyer matches

The buyer opens the order and supplies the **receive address** for the released tokens. Matching commits the buyer to pay, so it is signed too:



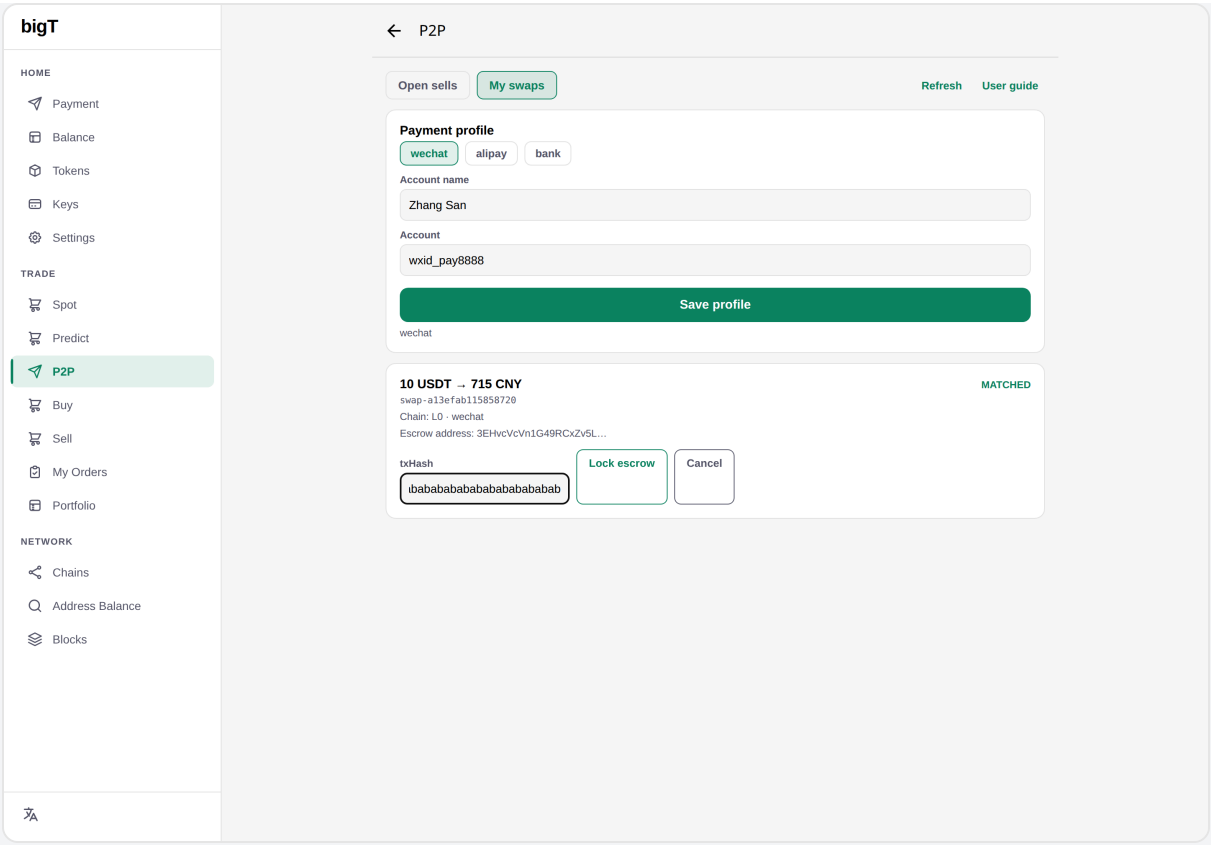
The engine creates a unique `swapId` for the lifecycle and the swap moves to `MATCHED`, shown on the buyer's **My swaps** tab:



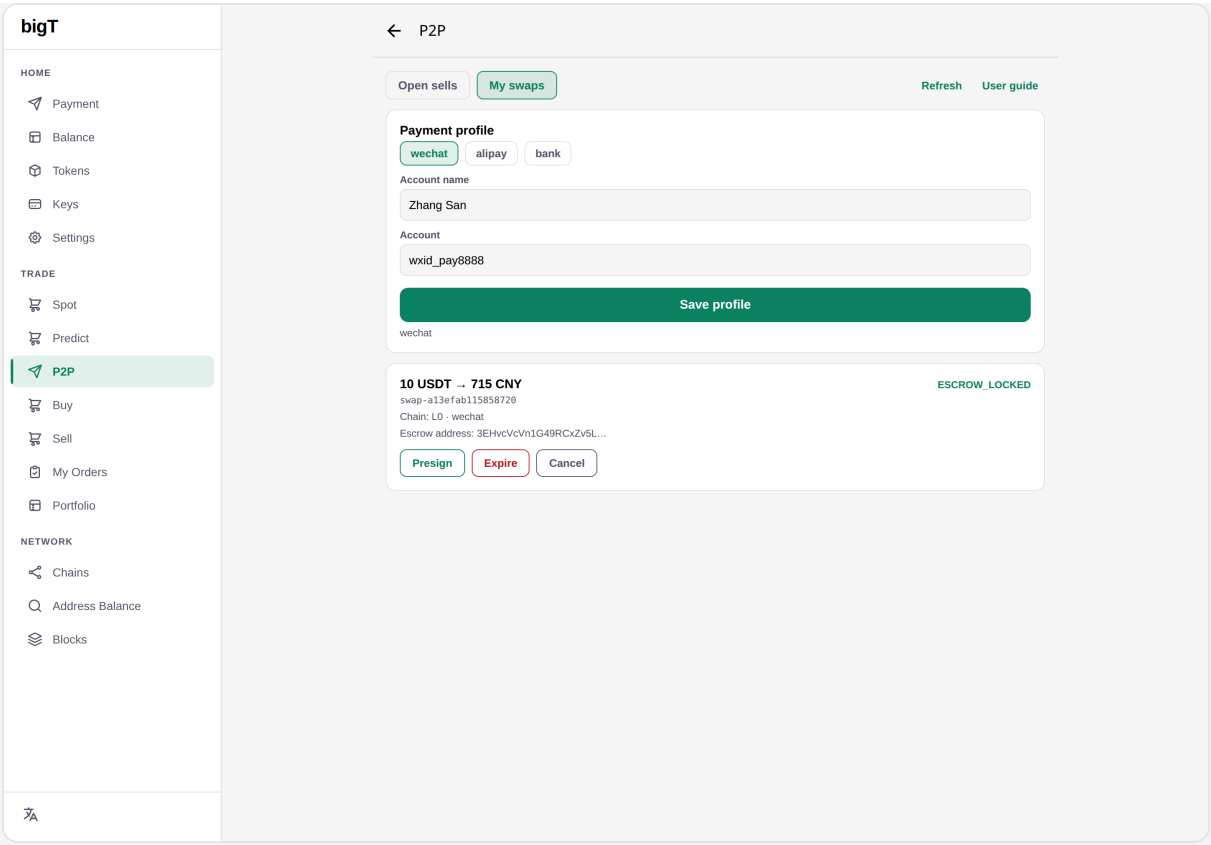
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4. Escrow lock — the seller funds L0

The seller sends the tokens to the deterministic 2-of-3 escrow address and reports the transfer's `txHash`. The engine proves the lock from the chain (`CONFIRMED`, destination `escrowAddress`, amount) and fails closed if any check does not hold:



The swap is now `ESCROW_LOCKED`. The buyer sees the funds secured **before** paying any fiat, and gets a **Payment details** action:



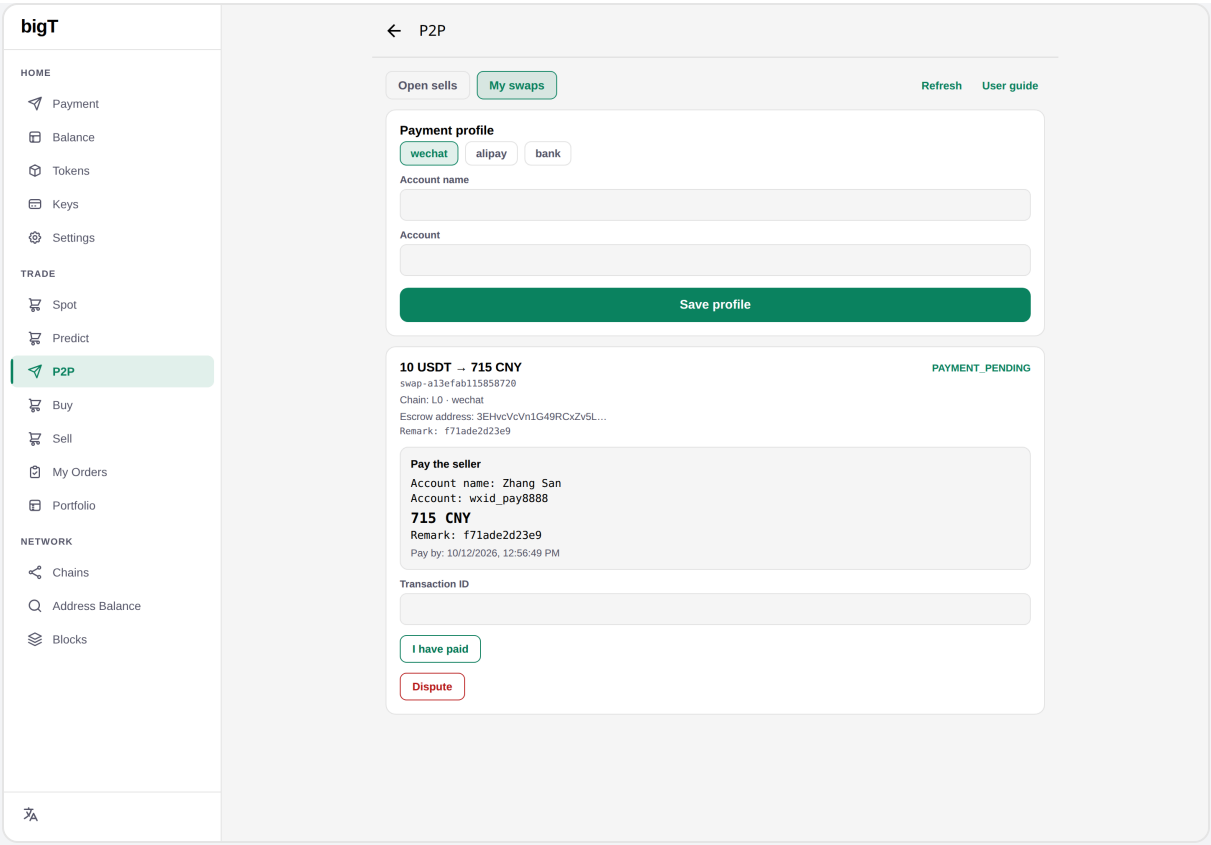
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5. The buyer pulls the payment instructions

The buyer presses **Payment details**; the engine reveals the seller's profile

and mints a **per-swap remark code** the transfer *must* carry, plus the exact

amount and a deadline. The instructions render inline in the swap card:

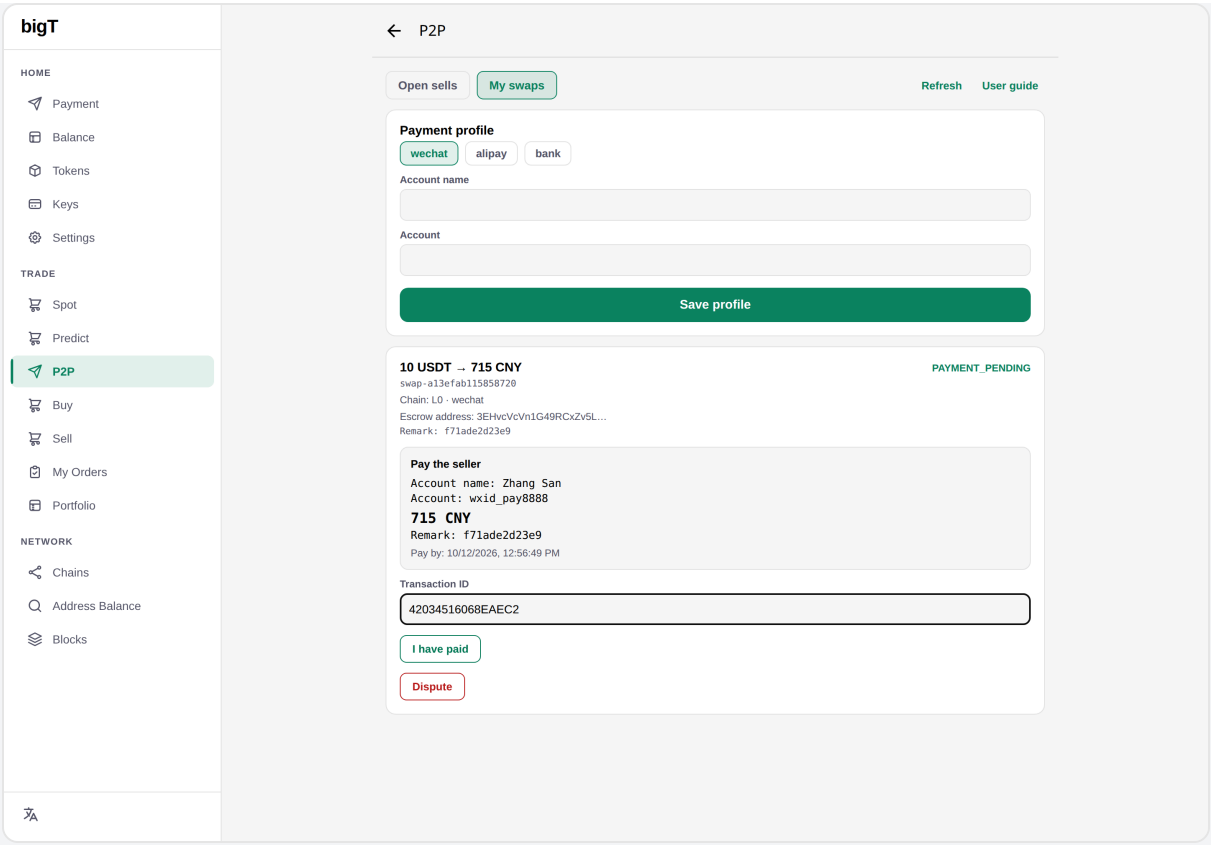


The swap is `PAYMENT_PENDING`, and a timeout starts — if the buyer never pays, the seller can expire and refund the escrow without the buyer's consent.

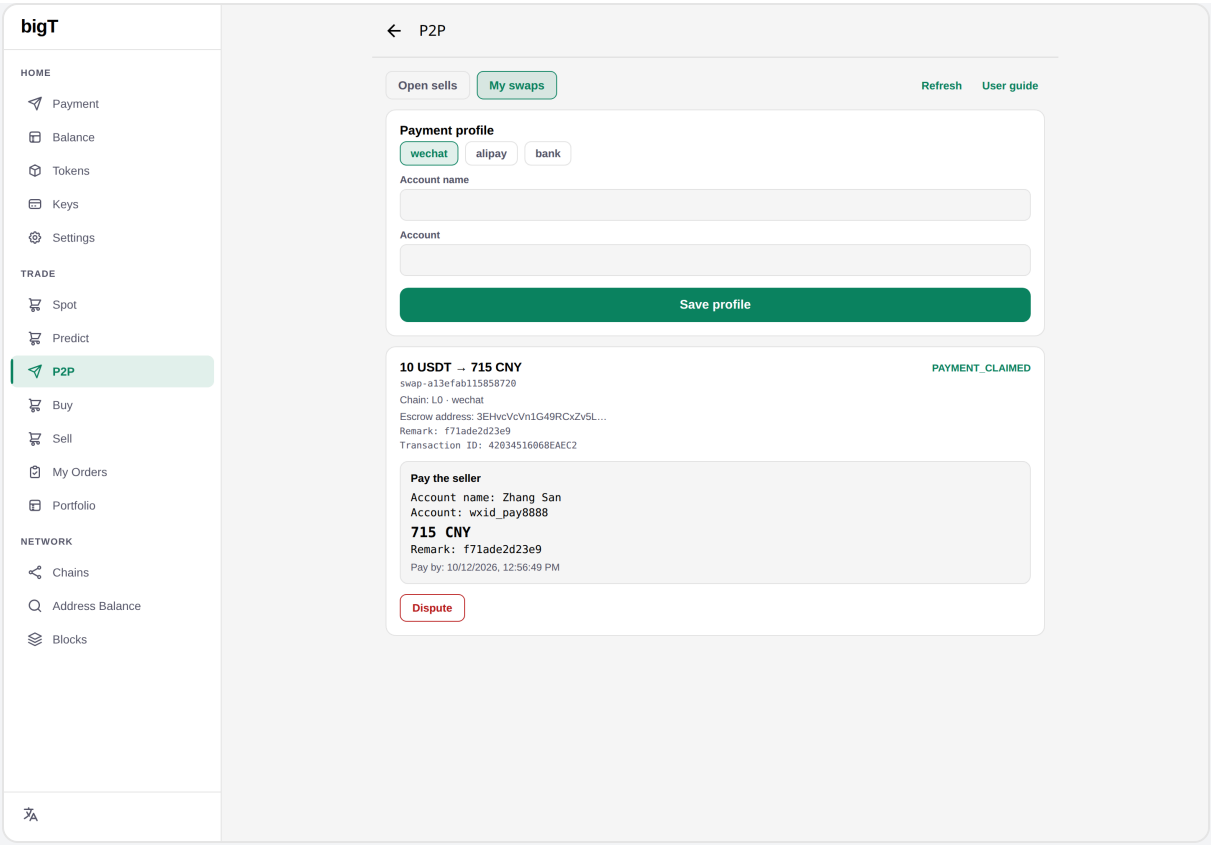
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6. The buyer pays and claims with the 流水号

The buyer transfers the exact amount in their own WeChat/Alipay/bank app, pastes the transaction number (流水号) into **Transaction ID**, and presses **I have paid**. An optional receipt image is stored server-side; only its `sha256` is anchored on the swap event:



Status is now PAYMENT_CLAIMED — **the tokens are still locked.** This state exists purely as the buyer's claim:



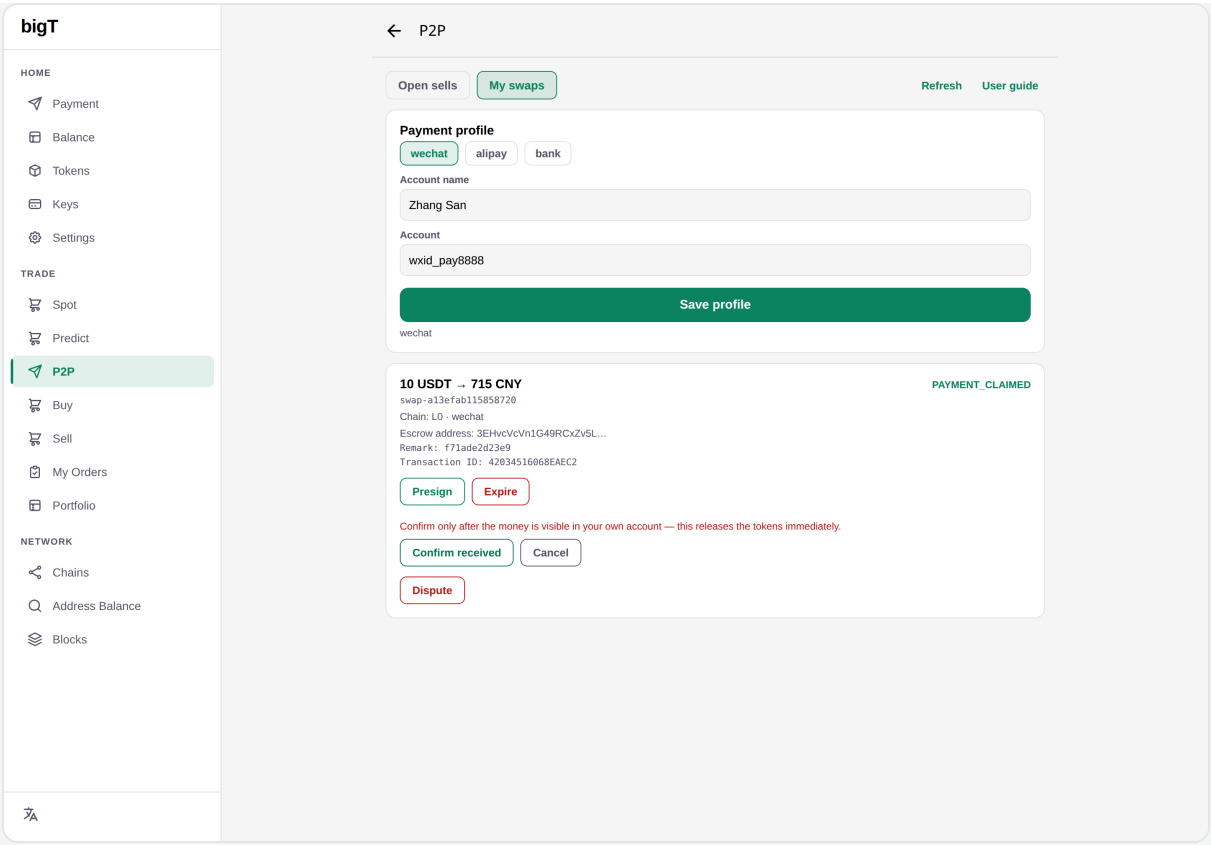
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7. The seller confirms on their own statement

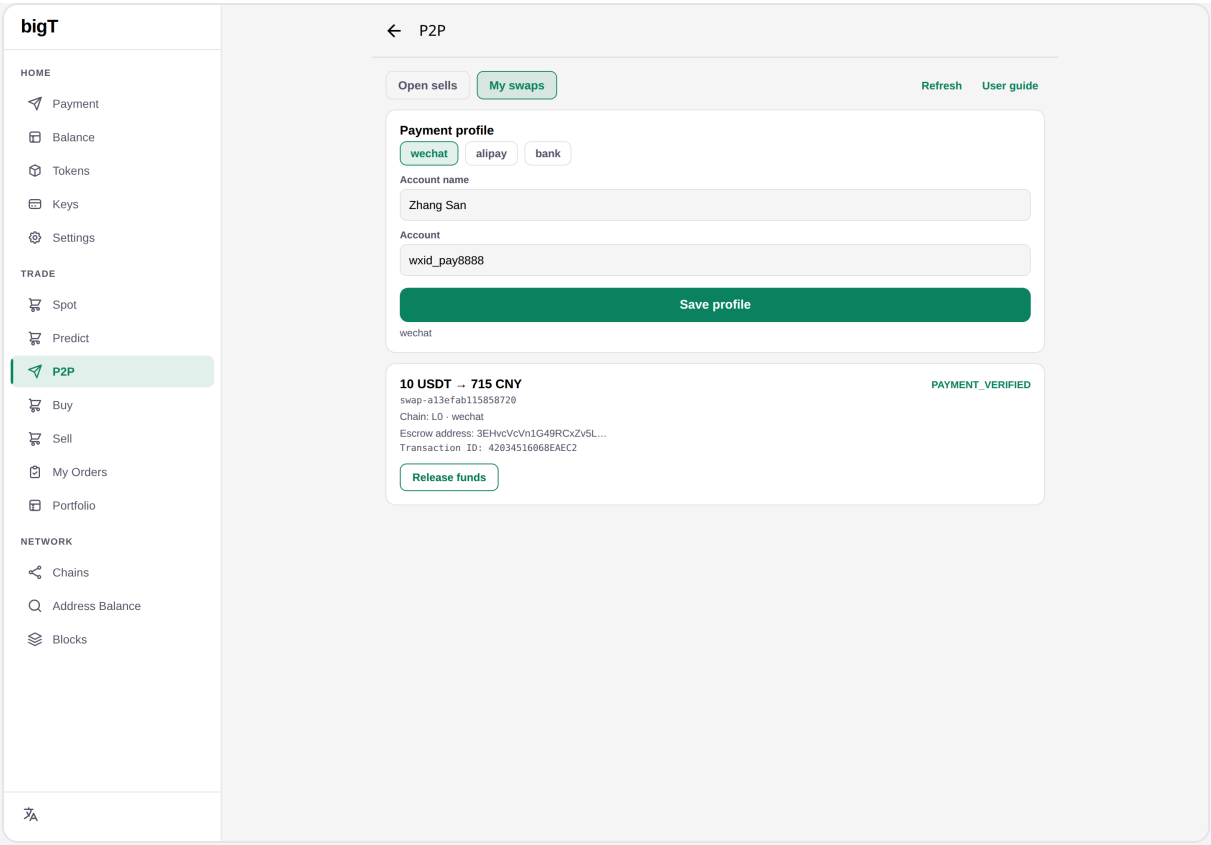
The seller's card shows the buyer's 流水号 and the remark. The seller checks

amount + remark + time + 流水号 in their own account and only then presses

Confirm received:



The swap is PAYMENT_VERIFIED :



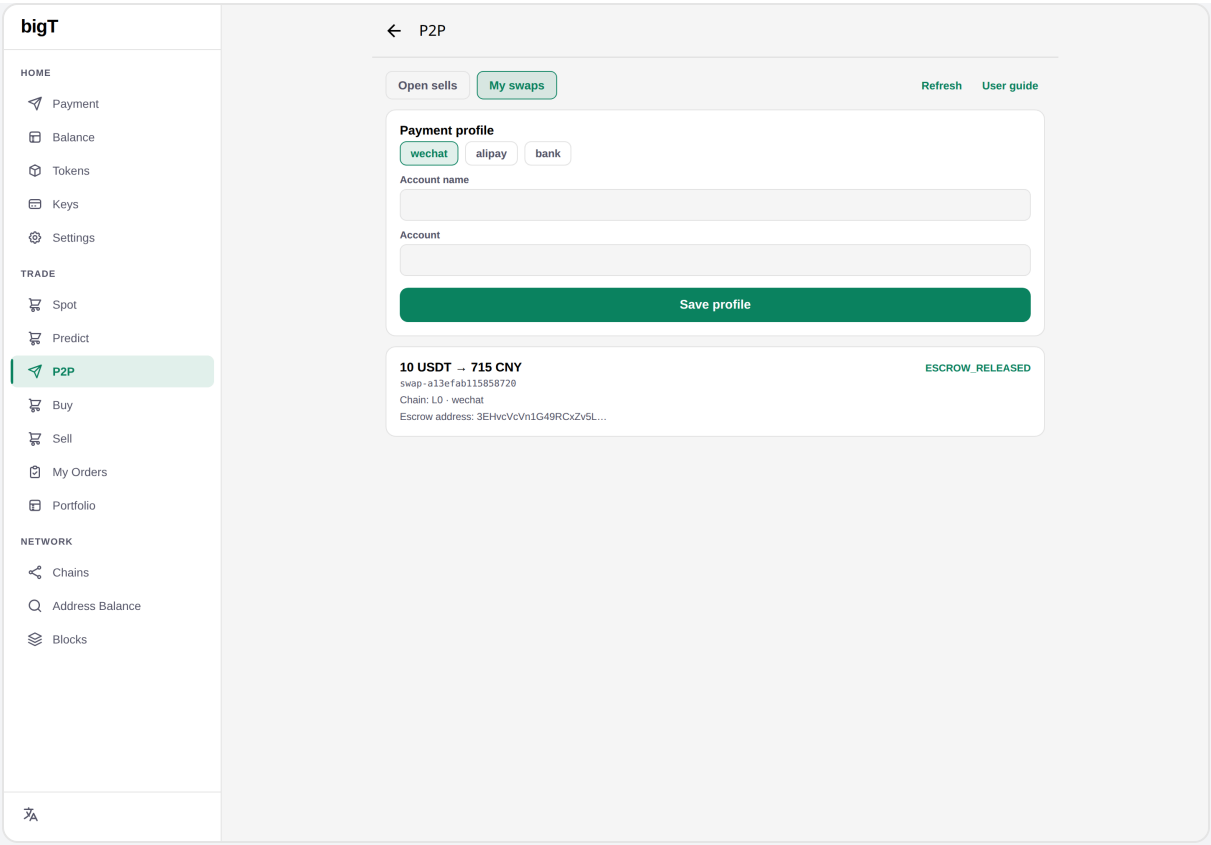
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8. Release — the funds move

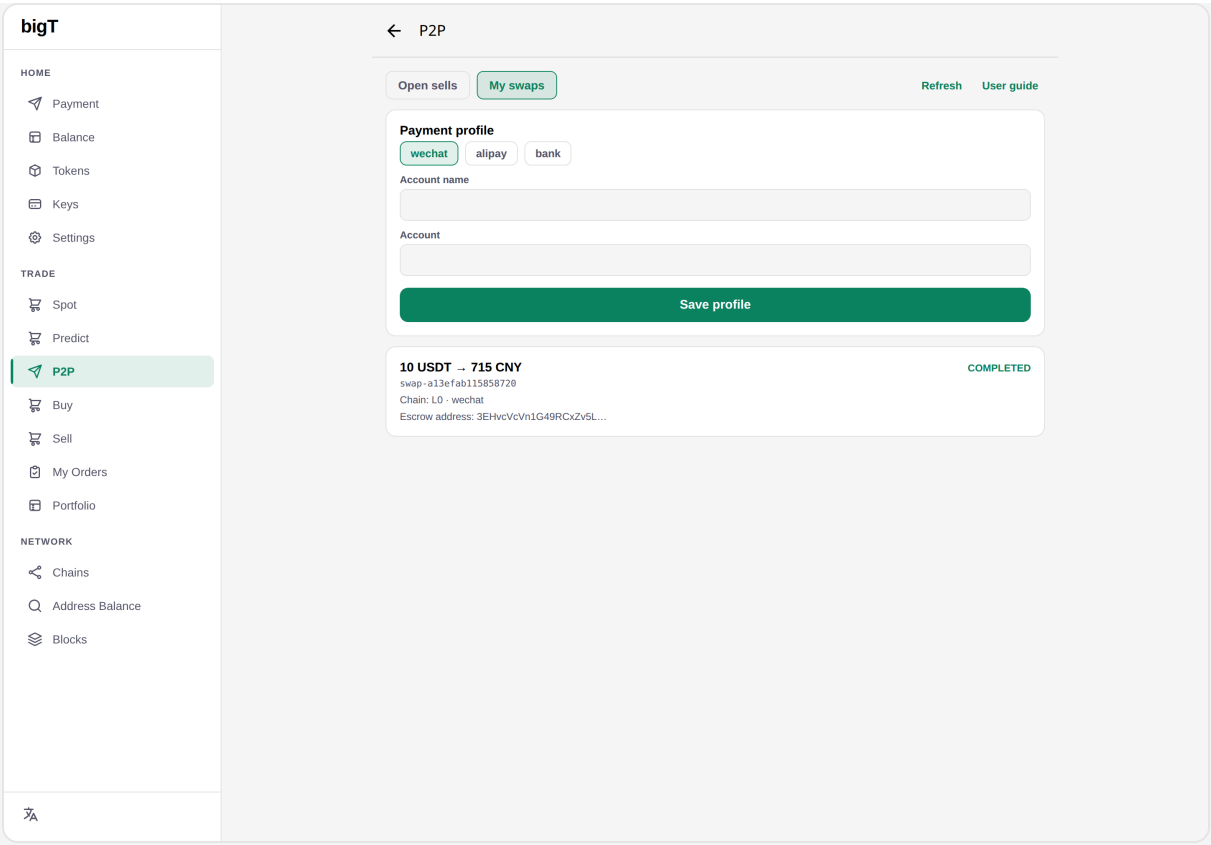
Release moves the escrowed tokens: the seller (presign or co-sign) and the

engine each sign the spend from the escrow output, and two signatures satisfy the

2-of-3 script. The tokens land at the buyer's receive address:



Because the seller was paid directly, there is no fiat payout step — the swap completes:



What protects you

Risk	Mitigation
Counterparty walks away	Funds sit in a 2-of-3 P2SH address; nobody can move them alone
Under- or over-payment	The amount is fixed at match; instructions carry the exact figure and a per-swap remark
Fake payment claim	The buyer's claim alone never releases — the seller confirms on their own statement
Seller claims non-receipt	Arbitration; the chain record and the proof store are the evidence
Engine goes rogue	Engine-only steps require the engine DID signature; every transition is anchored and publicly auditable

Dispute	Either party freezes the swap; the tokens stay locked until it resolves
Refund	`expire → refund`, co-signed by the engine and the seller — no buyer consent, no timelock

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Going live

The demo runs against the in-repo settlement service with a memory store and no chain check. The real flow wires the escrow signer to the L0 broadcast step and runs the engine against L1-SOCIAL for the order book and the audit anchor; the buyer pays the seller directly, and the engine still only ever co-signs.